

COST VERSUS POWER IN LIGHTING

And Some Suitable Solutions



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Ever wondered about the size of a salt grain? A single salt grain measures about three mil, which is a thousandth of an inch. Now, imagine a third of that size, placed on a printed circuit board (PCB). That is the size of the smallest LED available today. People do not regularly go for it, but it is heartening to know that something that small can also brighten something up.

That is where the surface-mount device light-emitting diode (SMD LED) module comes into the picture. Now, you have a $0.65\text{mm} \times 0.35\text{mm} \times 0.2\text{mm}$ -size LED on a PCB that can be used for a range of different purposes. Wide viewing angles of about 145° and low power consumption add to the charm.

SMD is going strong

An SMD chip is created from layers of man-made nano-sapphire and gallium crystal substrates. After growing these crystals, these are sliced, layered and bonded to a ceramic base. The chip can be then mounted in the required application. Applications of an SMD include everything from light bulbs to strip lights. One interesting use case would be the indicators on mobile phones.

Over the years, SMD LED has continued to improve its luminescence in accordance with Haitz law. Unlike typical LEDs, SMD chips can have two, four or six contacts,

COBs are easier to place on a board. The pins in SMD require precision soldering



Haitz's law

For a given wavelength of light, cost per lumen falls by a factor of 10, while the amount of light generated per LED package increases by a factor of 20, every decade.

depending on the number of diodes on the chip that may produce multiple colours. If a chip includes red, green and blue diodes, you have the possibility of creating any colour you want. This has resulted in SMDs becoming the prime choice for DC power applications, mostly due to the ease of use.

COB is another option

A COB LED has great advantages of thermal resistance, larger cooling area, better lighting effect and high light efficacy. It is basically an LED built directly on the circuit board. P. Chow Reddy, managing director, Interleaved Technologies, explains, "A COB contains multiple chips within one large wafer. This allows a larger light output and a more even spread of light." It allows connection of the bare chip to the substrate by conductive or non-conductive adhesive and wire bonding to achieve its electrical connection. Due to fewer components in the LED chip, thermal conductivity is directed at the circuit board and without the traditional LED chip packaging structure.

Heat emissions from LED chips are on average 25 per cent lower as compared to conventional LEDs, resulting in high efficiency and lower failure rates. And since the chip is built directly onto the circuit board, it reduces the need for spot soldering. After all, fewer soldering points mean less chances of failure.

Need for traditional LED packaging parts and PC lenses has also reduced, resulting in an increase in light visibility range and a decrease in light loss from lenses. COB LEDs are generally larger than SMDs and range from a few millimetres to a few centimetres-